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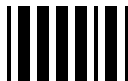
Baxter

Cernevit

Presentation

Each vial contains:

Retinol palmitate	
corresponding to retinol	3500 IU
Cholecalciferol	220 IU
DL α -tocopherol	10.200 mg
corresponding to α -tocopherol	11.200 IU
Ascorbic acid	125.000 mg
Coccarboxylase tetrahydrate	5.800 mg
corresponding to thiamine	3.510 mg
Riboflavin dihydrated sodium phosphate	5.670 mg
corresponding to riboflavin	4.140 mg
Pyridoxine hydrochloride	5.500 mg
corresponding to pyridoxine	4.530 mg
Cyanocobalamin	0.006 mg
Folic acid	0.414 mg
Dexpantenol	16.150 mg
corresponding to pantothenic acid	17.250 mg
D-Biotin	0.069 mg
Nicotinamide	46.000 mg
<i>Other ingredients</i>	
Glycine	250.000 mg
Glycocholic acid	140.000 mg
Soybean phosphatides	112.500 mg
Sodium hydroxide	} qs pH 5.9
Hydrochloric acid	



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Pharmaceutical form and Description

Powder for solution for injection or infusion, orange yellow sterile cake of powder.

Uses

CERNEVIT is indicated for adults and children over 11 years of age. When the daily requirements of vitamins are required to be given to the patient by the parenteral route because oral administration is either contraindicated, impossible or insufficient (eg due to malnutrition, gastrointestinal malabsorption, etc).

Dosage and administration

Dosage

Adults and children aged over 11 years : 1 vial/day.

Administration

Method of reconstitution: see section Instruction for use and handling.

Intravenous route: By slow intravenous injection (at least 10 minutes) or by infusion in a solution of 5% glucose or 0.9% sodium chloride solution for infusion.

Cernevit may be included in the composition of nutritive mixtures combining carbohydrates, lipids, amino acids and electrolytes provided that compatibility and stability have been confirmed for each nutritive mixture, to meet nutrient needs and prevent deficiencies and complications from developing.

The total vitamin amounts from all sources such as nutritional sources, other vitamin supplements, or medications that contain vitamins as inactive ingredients (see Interaction with Other Medicaments and Other Forms of Interaction) should be considered.

The patient's clinical status and vitamin levels should be monitored to ensure maintenance of adequate levels.

It should be taken into account that some vitamins, especially A, B2, and B6 are sensitive to ultraviolet light (e.g., direct or indirect sun light). In addition,

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loss of vitamins A, B1, C, and E may increase with higher levels of oxygen in the solution. These factors should be considered if adequate vitamin levels are not achieved.

Contraindications

- hypersensitivity to the active substances, especially vitamin B1 or to any of the excipients, including soy protein/products (lecithin in mixed micelle is soy-derived) or peanut protein/products
- hypervitaminosis from any vitamin contained in this formulation
- neonates, infants and children less than 11 years old.

Special Warnings and Precautions for Use

WARNINGS

Hypersensitivity Reactions

- Severe systemic hypersensitivity reactions have been reported with Cernevit, other multivitamin preparations, and individual vitamins (including B1, B2, B12 and folic acid). Reactions with fatal outcome have been reported with Cernevit and other parenteral vitamin products (See Undesirable Effects).
- Cyanocobalamin contains cobalt and may cause sensitivity reactions in individuals with cobalt allergy.
- Cross-allergic reactions between soybean and peanut proteins have been observed.
- In some cases, the manifestations of a hypersensitivity reaction during intravenous administration of multivitamins may be rate related. If infused intravenously, Cernevit should be administered slowly. If injected intravenously, the injection must be administered slowly (over at least 10 minutes).
- The infusion or injection must be stopped immediately if signs or symptoms of a hypersensitivity reaction develop.
- Mild allergic reactions such as sneezing or mild asthma are warning signs that a further injection may give rise to anaphylactic shock.

Vitamin Toxicity

- The patient's clinical status and blood vitamin concentrations should be

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monitored to avoid overdose and toxic effects, especially with vitamins A, D and E, and in particular in patients who receive additional vitamins from other sources or use other agents that increase the risk of vitamin toxicity.

- Monitoring is particularly important in patients receiving long-term supplementation.

Hypervitaminosis A

- The risk for hypervitaminosis A and vitamin A toxicity (e.g., skin and bone abnormalities, diplopia, cirrhosis) is increased in, for example:
 - patients with protein malnutrition,
 - patients with renal impairment (even in the absence of vitamin A supplementation),
 - patients with hepatic impairment,
 - patients with small body size (e.g., paediatric patients), and
 - patients on chronic therapy.
- Acute hepatic disease in patients with saturated hepatic vitamin A stores can lead to the manifestation of vitamin A toxicity.

Refeeding Syndrome in Patients Receiving Parenteral Nutrition

Refeeding severely undernourished patients may result in refeeding syndrome that is characterized by the shift of potassium, phosphorus, and magnesium intracellularly as the patient becomes anabolic. Thiamine deficiency and fluid retention may also develop. Careful monitoring and slowly increasing nutrient intakes while avoiding overfeeding can prevent these complications. Should nutrient deficiencies occur, appropriate supplementation may be warranted.

Precipitates in Patients Receiving Parenteral Nutrition

Pulmonary vascular precipitates have been reported in patients receiving parenteral nutrition. In some cases, fatal outcomes have occurred. Excessive addition of calcium and phosphate increases the risk of the formation of calcium phosphate precipitates. Precipitates have been reported even in the absence of phosphate salt in the solution. Precipitation distal to the in-line filter and suspected precipitate formation in the blood stream have also been reported.

In addition to inspection of the solution, the infusion set and catheter should also periodically be checked for precipitates.

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If signs of pulmonary distress occur, the infusion should be stopped and medical evaluation initiated.

PRECAUTIONS

Hepatic Effects

- Monitoring of liver function parameters is recommended in patients receiving Cernevit. Particularly close monitoring is recommended in patients with hepatic jaundice or other evidence of cholestasis. In patients receiving Cernevit, instances of liver enzyme increases have been reported, including isolated alanine aminotransferase (ALT) increases in patients with inflammatory bowel disease (see Undesirable Effects). In addition, an increase in bile acid levels (total and individual bile acids including glycocholic acid) have been reported in patients receiving Cernevit.
- Hepatobiliary disorders including cholestasis, hepatic steatosis, fibrosis and cirrhosis, possibly leading to hepatic failure, as well as cholecystitis and cholelithiasis are known to develop in some patients on parenteral nutrition (including vitamin supplemented parenteral nutrition). The etiology of these disorders is thought to be multifactorial and may differ between patients. Patients developing abnormal laboratory parameters or other signs of hepatobiliary disorders should be assessed early by a clinician knowledgeable in liver diseases in order to identify possible causative and contributory factors, and possible therapeutic and prophylactic interventions.

Use in Patients with Impaired Hepatic Function

Patients with hepatic impairment may need individualized vitamin supplementation. Particular attention should be placed on preventing vitamin A toxicity, because the presence of liver disease is associated with increased susceptibility to vitamin A toxicity, in particular in combination with chronic excessive alcohol consumption (See also Hypervitaminosis A and Hepatic Effects above).

Use in Patients with Impaired Renal Function

Patients with renal impairment may need individualized vitamin

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supplementation, depending on the degree of renal impairment and the presence of concomitant medical conditions. In patients with severe renal impairment, particular attention should be placed on maintaining adequate vitamin D status and preventing vitamin A toxicity, which may develop in such patients with low-dose vitamin A supplementation or even without supplementation.

Pyridoxine (vitamin B6) hypervitaminosis and toxicity (peripheral neuropathy, involuntary movements) have been reported in patients on chronic haemodialysis receiving intravenous multivitamins containing 4 mg pyridoxine administered three times a week.

General Monitoring

Clinical status and vitamin levels should be monitored in patients receiving parenteral multivitamins as the only source of vitamins for extended periods of time. It is particularly important to monitor for adequate supplementation of, for example:

- Vitamin A in patients with pressure ulcers, wounds, burns, short bowel syndrome or cystic fibrosis
- Vitamin B1 in dialysis patients
- Vitamin B2 in cancer patients
- Vitamin B6 in patients with renal impairment
- Individual vitamins whose requirements may be increased due to interactions with other medicines (see Interaction with Other Medicaments and Other Forms of Interaction)

Deficiency of one or more vitamins must be corrected by specific supplementation.

Vitamin K

Cernevit does not contain Vitamin K. Vitamin K must be administered separately if necessary.

Use in Patients with Vitamin B12 Deficiency

Evaluation of vitamin B12 status is recommended before starting supplementation with Cernevit in patients at risk for vitamin B12 deficiency and/or when supplementation with Cernevit over several weeks is planned.

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After several days of administration, both the individual amounts of cyanocobalamin (vitamin B12) and folic acid in Cernevit may be sufficient to result in an increase in red blood cell count, reticulocyte count, and haemoglobin values in some patients with vitamin B12 deficiency-associated megaloblastic anaemia. This may be masking an existing vitamin B12 deficiency. Effective treatment of vitamin B12 deficiency requires higher doses of cyanocobalamin than provided in Cernevit.

Folic acid supplementation in patients with vitamin B12 deficiency, who do not also receive vitamin B12, does not prevent the development or progression of neurologic manifestations associated with the vitamin B12 deficiency. It has been suggested that neurologic deterioration may even be accelerated.

When interpreting levels of vitamin B12, it should be taken into account that recent intake of vitamin B12 may result in normal levels despite a tissue deficiency.

Laboratory Test Interferences

Biotin

Biotin may interfere with laboratory tests that are based on a biotin/streptavidin interaction, leading to either falsely decreased or falsely increased test results, depending on the assay. The risk of interference is higher in children and patients with renal impairment and increases with higher doses. When interpreting results of laboratory tests, possible biotin interference has to be taken into consideration, especially if a lack of coherence with the clinical presentation is observed (e.g. thyroid test results mimicking Graves' disease in asymptomatic patients taking biotin or false negative troponin test results in patients with myocardial infarction taking biotin). Alternative tests not susceptible to biotin interference should be used, if available, in cases where interference is suspected. The laboratory personnel should be consulted when ordering laboratory tests in patients taking biotin.

Ascorbic acid

Depending on the reagents used, the presence of ascorbic acid in blood and

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urine may cause false high or low glucose readings in some urine and blood glucose testing systems, including test strips and handheld glucose meters. The technical information for any laboratory test should be consulted to determine the potential interference from vitamins.

Sodium Content

Cernevit contains 24 mg sodium (1 mmolL) per vial. This should be taken into consideration if patients are on a controlled sodium diet.

Paediatric Use

Cernevit is indicated in paediatric patients over 11 years of age (see also Hypervitaminosis A above).

Geriatric Use

In general, dosage adjustments for an elderly patient should be considered (reducing the dose and/or extending the dosing intervals) reflecting the greater frequency of decreased hepatic, renal, or cardiac function, and of concomitant disease or drug therapy.

Fertility, pregnancy and lactation

Physicians should carefully consider the potential risks and benefits for each specific patient before prescribing CERNEVIT.

No safety data from Clinical trials are available to date therefore it is recommended to use CERNEVIT during pregnancy and breastfeeding only if absolutely necessary (only if the potential benefits are higher than the foetal risks).

CERNEVIT contains vitamin A. Taking into account that normal nutrition covers the daily vitamin A needs (found in liver, liver products, milk, dairy products, margarine, eggs, salad oils), daily doses sometimes exceed the daily needs (especially when liver or liver products are consumed). Certain data show that high vitamin A doses in pregnant women (1st trimester) and women who want to become pregnant should not exceed 10 000 IU/day. This should be taken into consideration when CERNEVIT is administered with a large amount of food containing vitamin A.

The risk of vitamin A overdose in the neonate should be evaluated before

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using CERNEVIT in breastfeeding women.

Fertility

There are no adequate data from the use of CERNEVIT with regards to fertility in male or female patients.

Interaction with Other Medicaments and Other Forms of Interaction

Interactions between specific vitamins in Cernevit and other agents should be managed accordingly.

Such interactions include:

- Agents that can cause pseudotumor cerebri (including certain tetracyclines): Increased risk for pseudotumor cerebri by concomitant administration of Vitamin A
- Alcohol (chronic excessive consumption): Increases the risk of vitamin A hepatotoxicity
- Anticonvulsants (phenytoin, fosphenytoin, phenobarbital, primidone): Folic acid supplementation can decrease the anticonvulsant serum concentration and increase seizure risk.
- Antiplatelet agents (e.g., aspirin): Vitamin E can add to the inhibition of platelet function
- Aspirin (high dose therapy): Can reduce folic acid levels by increasing urinary excretion
- Certain anticonvulsants (e.g., phenytoin, carbamazepine, phenobarbital, valproate): Can cause folate, pyridoxine and vitamin D deficiencies
- Certain antiretroviral agents: Decreased vitamin D levels have been associated with, e.g., efavirenz and zidovudine. Decreased formation of the active vitamin D metabolite has been associated with protease inhibitors.
- Chloramphenicol: Can inhibit the haematological response to vitamin B12 therapy
- Deferoxamine: Increased risk of iron-induced cardiac failure due to increased iron mobilization by supraphysiologic vitamin C supplementation. For specific precautions, refer to deferoxamine product information.
- Ethionamide: Can cause pyridoxine deficiency

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- Fluoropyrimidines (5-fluorouracil, capecitabine, tegafur): Increased cytotoxicity when combined with folic acid
- Folate antagonists, e.g., methotrexate, sulfasalazine, pyrimethamine, triamterene, trimethoprim, and high doses of tea catechins: Block the conversion of folate to its active metabolites and reduce the effectiveness of supplementation
- Folate antimetabolites (methotrexate, raltitrexed): Folic acid supplementation can decrease the antimetabolite effects
- Levodopa: The content of pyridoxine may interfere with the effects of concurrent levodopa therapy.
- Pyridoxine antagonists, including cycloserine, hydralazine, isoniazid, penicillamine, phenelzine: Can cause pyridoxine deficiency
- Retinoids, including bexarotene: Increase the risk of toxicity when used concomitantly with vitamin A (see Hypervitaminosis A)
- Theophylline: Can cause pyridoxine deficiency
- Tipranavir oral solution: Contains 116 IU/mL of vitamin E, which is in excess of the daily recommended intake
- Vitamin K antagonists (e.g., warfarin): Enhanced anticoagulant effect by vitamin E

Drugs that Bind to alpha1-Acid Glycoprotein (AAG):

In an in vitro study using human serum, concentrations of glycocholic acid approximately 4 times higher than the glycocholic acid serum concentration that would result from a bolus injection of Cernevit in adults, increased the unbound fraction of selected drugs known to bind to alpha1-acid glycoprotein (AAG) by 50- 80%.

It is not known whether this effect is clinically relevant if the amount of glycocholic acid contained in a standard Cernevit dose (as a component of the mixed micelles) is administered by slow intravenous injection, intramuscular injection, or infused over a longer period of time.

Patients receiving Cernevit as well as drugs that bind to AAG should be closely monitored for increases in response of these drugs. These include

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propranolol, prazosin, and numerous others.

Interactions with Additional Vitamin Supplementation:

Some medications can interact with certain vitamins at doses markedly higher than those provided with Cernevit. This should be taken into consideration in patients receiving vitamins from multiple sources, and when applicable, patients should be monitored for such interactions and managed accordingly.

Such interactions include:

- Amiodarone: Concomitant use of vitamin B6 can enhance amiodarone-induced photosensitivity.
- Agents with anticoagulant effects (e.g., such as abciximab, clopidogrel, heparin, warfarin): Increased bleeding risk due to additional risk of bleeding associated with high vitamin A doses
- Carbamazepine: Inhibition of metabolism associated with large nicotinamide doses
- Chemotherapeutic agents that rely on the production of reactive oxygen species for their activity: Possible inhibition of chemotherapy activity by the antioxidant effects of high doses of vitamin E
- Insulin, antidiabetic agents: Decreased insulin sensitivity associated with large nicotinamide doses
- Iron: High dose-supplementation with vitamin E may reduce the haematological response to iron in anaemic patients
- Oral contraceptives (combination hormone types): High doses of vitamin C have been associated with breakthrough bleeding and contraceptive failure
- Phenobarbital: Increased metabolism/lower serum levels and reduced effect associated with large pyridoxine doses
- Phenytoin, fosphenytoin: Lower serum levels associated with large pyridoxine doses
- Primidone: Decreased metabolism to phenobarbital and increased primidone levels associated with large nicotinamide doses

Pernicious anaemia

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The folic acid in Cernevit may obscure pernicious anaemia.

Effects on the ability to drive and use machines

There is no information on the effects of Cernevit on the ability to operate an automobile or other heavy machinery.

Undesirable Effects

Adverse drug reactions (ADRs) that occurred after administration of Cernevit are presented with their relative frequencies; these include ADRs documented in clinical trials and those from post-marketing reports. Cernevit was administered during 3 clinical trials to 267 adult patients requiring a parenteral vitamin supplement.

Frequencies of ARs are reported, using the following convention: very common (≥1/10); common (≥1/100 to <1/10); uncommon (≥1/1000 to <1/100); rare (≥1/10000 to <1/1000); very rare (<1/10000); and unknown (cannot be estimated from the available data).

Clinical Trial and Post-Marketing Adverse Drug Reactions Reported for Cernevit:

System Organ Class	Preferred MedDRA Term	Frequency ¹
Immune system disorders	Systemic hypersensitivity reactions with manifestations such as respiratory distress, chest discomfort, throat tightness, urticaria, rash, erythema, epigastric discomfort, as well as cardiac arrest with fatal outcome	Unknown
Metabolism and nutrition disorders	Vitamin A increased ^{2,3} , Retinol binding protein increased ³	Unknown Unknown
Nervous system disorders	Dysgeusia (metallic taste)	Unknown
Cardiac disorders	Tachycardia	Unknown
Respiratory, thoracic and mediastinal disorders	Tachypnea	Unknown

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Gastrointestinal disorders	Nausea Vomiting Diarrhoea	Uncommon Uncommon Unknown
Hepatobiliary disorders	Transaminases increased, Isolated alanine aminotransferase increased ⁴ ,	Unknown Unknown
	Glutamate dehydrogenase increased, Blood alkaline phosphatase increased, Bile acids increased ⁵ Gamma-glutamyltransferase increased	Unknown Unknown Unknown Unknown
Skin and subcutaneous tissue disorders	Pruritus	Unknown
General disorders and administration site conditions	Injection/Infusion Site Pain	Common
	Pyrexia, Generalized aching, infusion site reactions, i.e., burning sensation, rash	Unknown Unknown Unknown Unknown

¹ The frequency either cannot be determined or the overall number of patients in the individual studies is too small to permit a valid estimation of frequency.

² No symptoms of hypervitaminosis A were reported

³ Elevated plasma vitamin A levels have been reported in 8 of 20 patients receiving Cernevit in parenteral nutrition at day 45 of administration. From day 45 to day 90 of product administration the high values of vitamin A remained stable (maximum observed value of 3.6 µmol/L at day 90; normal values: 1 to 2.6 µmol/L). In addition, an average increase in retinol binding protein (RBP) was also identified. A maximum observed RBP value of 60 mg/L at day 90 (normal values: 30 to 50 mg/L), was reported.

⁴ Isolated alanine aminotransferase increases was reported in the presence of inflammatory bowel disease. Cernevit was administered by intravenous injection in the absence of parenteral nutrition.

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⁵ An increase in total and individual bile acids including glycocholic acid has been reported to develop early in the course of parenteral nutrition administration in patients receiving Cernevit.

Overdosage

Acute or chronic overdose of vitamins (in particular A, B6, D, and E) can cause symptomatic hypervitaminosis.

Clinical signs of acute overdose of vitamin A (doses exceeding 150,000 IU):

- Gastrointestinal disorders, headache, raised intracranial pressure (swollen fontanelle in infants), papilloedema, psychiatric disorders, irritability, or even convulsions, delayed generalized desquamation.

Clinical signs of chronic intoxication (prolonged vitamin A supplementation with supra-physiological doses in non-deficient subjects)

- Liver disorders, raised intracranial pressure, cortical hyperostosis of long bones and premature epiphyseal fusion, cephalitis, pruritus, vomiting, dryness of mucous membranes. The diagnosis is generally based on the presence of tender or painful subcutaneous swellings in the extremities of the limbs. X-rays demonstrate diaphyseal periosteal thickening of the ulna, fibula, clavicles and ribs. The risk of overdose is particularly high if a patient receives vitamins from multiple sources and overall supplementation of a vitamin does not match the patient's individual requirements, and in patients with increased susceptibility to hypervitaminosis.

Actions to be taken in the event of acute or chronic overdose

- Stop administration of CERNEVIT, reduce calcium intake, increase diuresis and rehydrate.

Pharmacological Properties

Pharmacodynamic properties

ADDITIVES FOR INTRAVENOUS SOLUTIONS/VITAMINS,
(ATC Code: B05XC)

CERNEVIT contains 9 hydrosoluble vitamins and 3 liposoluble vitamins, essential to the metabolism of adults and children aged over 11, except for vitamin K.

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The pharmacodynamic properties of CERNEVIT are those of the individual 12 vitamins in its composition.

Pharmacokinetic properties

Not available.

Preclinical safety data

Not available.

Packing

Box of 10 vials

Storage

Store below 25°C, keep container in the outer carton. Protect from heat and light. To be used only as directed by medical practitioner. Reconstituted solution should be kept refrigerated (2° – 8°C) for no more than 24 hours. The product after reconstitution should be used immediately. From microbiological point of view, the product should be used immediately after reconstitution.

Shelf life

2 years

Incompatibility statement

- In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products.
- Additives may be incompatible with parenteral nutrition containing Cernevit.
- If co-administration of drugs that are incompatible at the Y-site is necessary, administer via separate IV lines.
- Vitamin A and thiamine in Cernevit may react with bisulfites in parenteral nutrition solutions (e.g., as a result of admixtures) leading to degradation of vitamin A and thiamine.
- An increase in pH of a solution may increase the degradation of some

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vitamins. This should be considered when adding alkaline solutions to the admixture containing Cernevit.

- Folic acid stability can be impaired with increased calcium concentrations in an admixture.

Instructions for Use/Handling

Aseptic conditions must be followed during reconstitution and when used as part of an admixture in parenteral nutrition.

Using a syringe, inject 5 ml of water for injections, or 5% glucose solution or 0.9% sodium chloride solution into the vial.

The obtained solution is yellow-orange in colour.

Mix gently to dissolve the lyophilized powder.

Before transfer from the vial, Cernevit must be completely dissolved.

Do not use unless the reconstituted solution is clear, the original seal is intact and the container is undamaged.

After addition of Cernevit to a parenteral nutrition solution, check for any abnormal colour change and/or the appearance of precipitates, insoluble complexes, or crystals.

Mix the final solution thoroughly when Cernevit is used as an admixture in parenteral nutrition.

Single use only. Any unused portion of reconstituted Cernevit should be discarded and should not be stored for subsequent admixing.

Use of a final filter is recommended during administration of all parenteral nutrition solutions.

Manufacturer

Baxter S.A.

Bd. René Branquart 80, B-7860 Lessines
Belgium

Date of last review

March 2024